

Quan A. Nguyen

(480) 862-4827 | Houston, TX, 77449 | quannnguyenanhnaq@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION & CERTIFICATION

Arizona State University Bachelor of Science in Computer Science

GPA: 3.7/4.0

Arizona State University Master of Science in Computer Engineering

GPA: 3.78/4.0

Relevant Coursework: Data Structures and Algorithms, Cloud Computing, Probability and Statistics, Computer Theory

EXPERIENCE

Software Developer Associate – AI

Oct 2025 – Present

Next Lab

Tempe, AZ

- Worked alongside a PhD candidate to develop state-of-the-art tiny LLMs with a local RAG pipeline on Raspberry Pi 5 (8GB, Ubuntu Linux), supporting offline semantic search and inference at ~6–18 tokens/sec with various quantization.
- Developed a Python-based ARM profiling framework to measure token throughput, memory usage, and CPU temperature to test inference of different models in different quantization (fp16, fp8, q4_k_m), which are guardrails to prevent crashes.
- Implemented vector retrieval and embedding indexing for long-context South African language datasets collected from 5+ verified data sources and applied labels and annotations by African-native volunteers for data accuracy and transparency.
- Upgraded Python, compiler toolchains, and library dependencies to support new LLM architectures on HPC NPU and CPU, achieving ~90% model size reduction by applying [Samsung Lab's LittleBit](#) method to allow extreme LLM compression.

Software Engineer

August 2024 – May 2025

Maricopa County Attorney Office

Phoenix, AZ

- Completed an application w/ a team of 5 developers, deploying a report-writing system using React.js and ASP.NET Core, backed by SQL Server database, and hosted with Windows Server to eliminate physical overhead for +30 active detectives.
- Reduced friction in report approval process by adding RBAC with REST APIs to streamline submissions and comments.
- Used EF Core for initial data migrations, created CI pipelines w/ Docker, and followed Agile Scrum practices which allowed the project to complete 2 weeks ahead of schedule, saving +\$800,000 for the county by not buying off-the-shelves products.

Cloud Developer & Teaching Assistant

August 2023 – May 2025

Amazon Web Services Cloud Innovation Center

Tempe, AZ

- Taught AWS services (S3, Lambda, EC2, load balancers, Kubernetes) through Zoom and hands-on labs for 150+ students.
- Provided guidance and workshops to understand use cases of AWS prior to taking [Cloud Practitioner Certificate exam](#).
- Automated grading workflows w/ AWS Lambda, serverless scheduling, and Canvas API, reduced workload by 95%.

Software Engineer Intern

June 2024 – August 2024

Crown Castle

Houston, TX

- Redesigned the fiber network search platform w/ Flask and [Neo4j](#), migrating from relational databases to graph databases.
- Optimized ETL pipelines processing 500K+ records by batching and indexing, improving query performance by 47%.
- Containerized and tested services w/ Docker and GitLab CI, integrating Prometheus and Grafana for performance monitoring, error tracking, and performed Scrum Meeting weekly for documentation on sprints and retrospectives.

Open-Source Contributions

Fill – Automated Loan Data Entry | JavaScript, Chrome Manifest V3, Agent Skills

- Wrote a [JavaScript extension](#) for Precious Realty & Mortgage, the largest Vietnamese-owned Real Estate firm (100 million USD), which communicates with the DOM to parse client's data and autofill loan estimates into a PDF.
- Completed and published within 2 weeks of development by developing targeted Agent Skills documents and Plans.

Medical Imaging ML Benchmark | Python, Tableau, SLURM

- Curated [custom "cookbooks"](#) for training deep learning models (UNet++, ConvNeXt, SWIN) for Chest X-Rays classification and segmentation tasks in supervised learning environment, up to date with current baselines.
- Utilized SLURM on High Performing Cluster to run large-scale trainings and Tableau for visualizations in conferences.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, C#, bash / shell, SQL, Cypher, TypeScript, HTML, CSS

Frameworks & Cloud: React.js, LangChain.js, Node.js, Neo4j, PostgreSQL, SQL Server, AWS, Microsoft Azure

Operating System & Tools: Linux, Unix, PowerShell, macOS, Git, GitHub, Docker, Postman, Microsoft Office

AI Automation: Claude Code, Cursor, Antigravity, Codex, MCP, Agent Skills/Rules